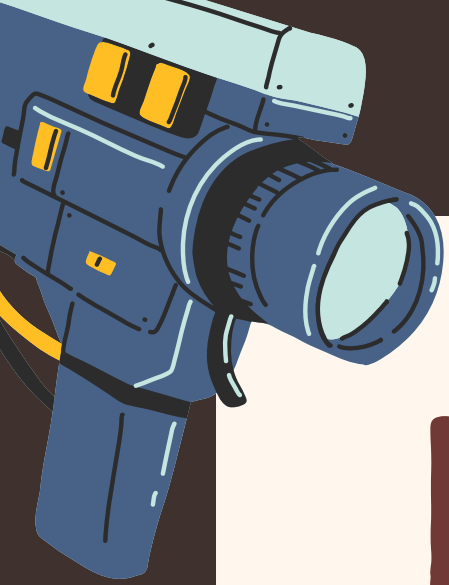




MOVIE SUCCESS PREDICTION

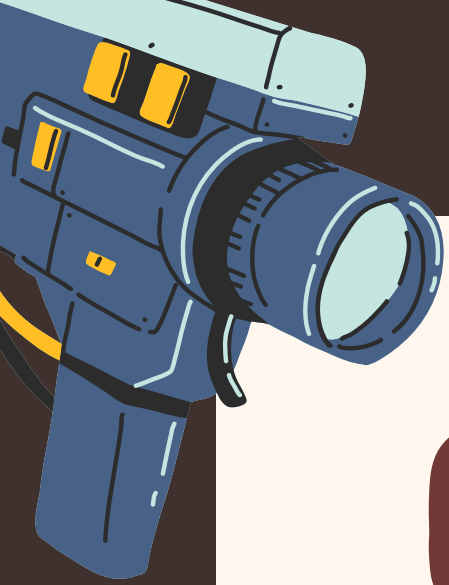
Presented By: [REDACTED], Annika
Rodriguez, and [REDACTED]



PROBLEM & MOTIVATION

- High financial risk in movie production
- Difficult to prediction box office success
- Data-driven approaches can improve decision-making

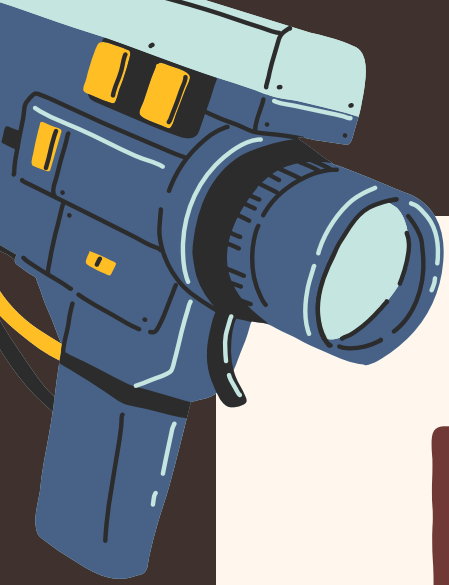




OBJECTIVES

- Predict whether a movie will be successful based on pre-release or available movie attributes
- Build a deep learning model for movie success classification
- Compare deep learning performance against baseline machine learning models
- Evaluate results using accuracy, precision, recall, and F1-score





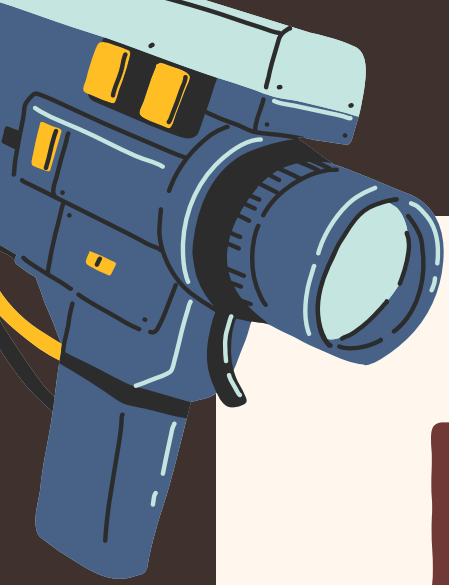
DATASET

- TMDB 5000 Movie Dataset
- ~5000 movies
- Includes financial and popularity features
- Key variables used:
 - Budget
 - Revenue
 - Popularity
 - Runtime



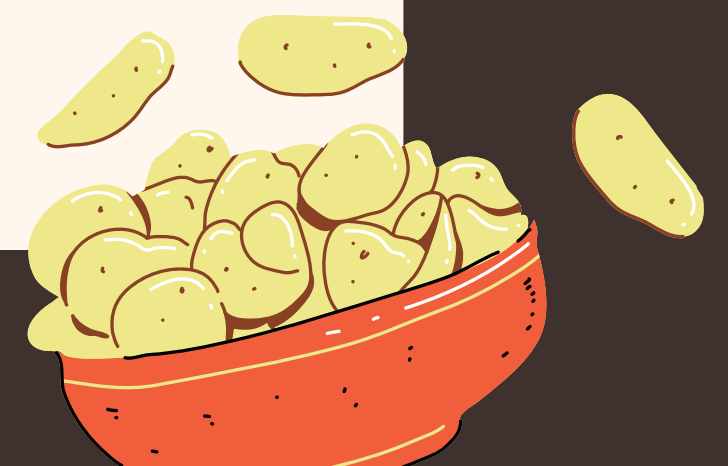
- Budget
- genres
- homepage
- id
- keywords
- original_language
- original_title
- overview
- popularity
- production_countries
- release_date
- revenue
- runtime
- spoken_languages
- status
- tagline
- title
- vote_average
- vote_count

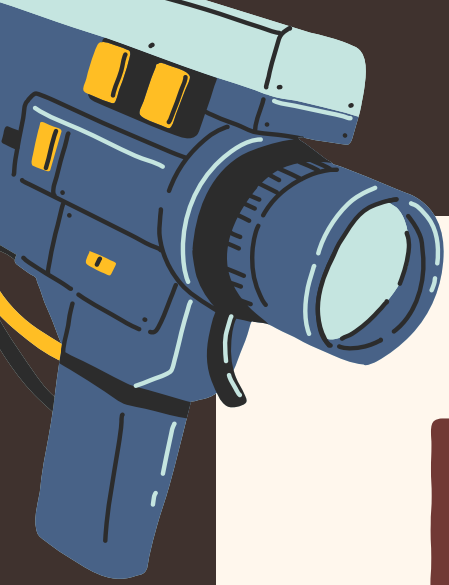




DATA PREPROCESSING

- Removed missing values
- Filtered invalid data:
 - Budget > 0
 - Revenue > 0
 - Runtime > 0
- Created target variable:
 - Success = revenue > median

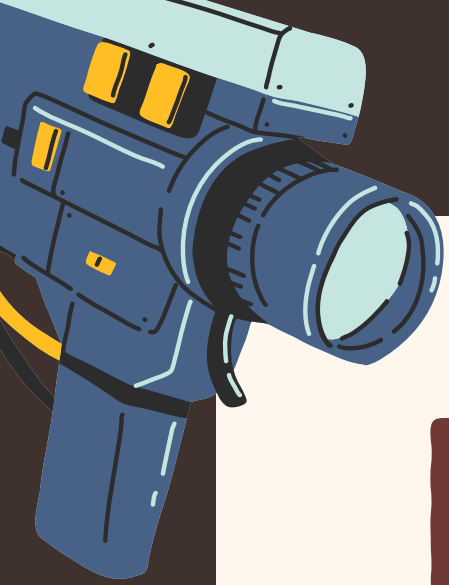




FEATURES & DATA SPLIT

- Features used:
 - Budget
 - Popularity
 - Runtime
 - Vote Average
- 80/20 Train-Test Split
- Standardized features using scaling

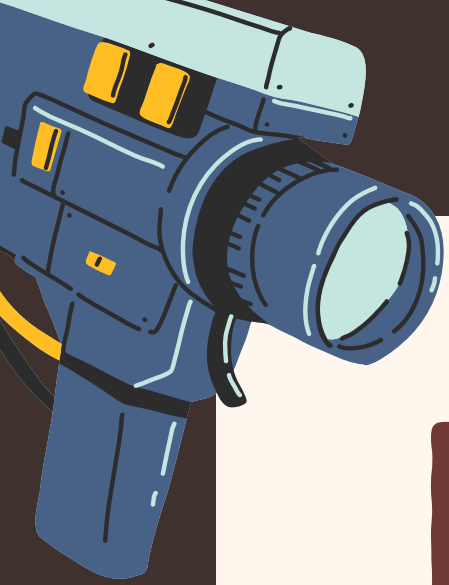




BASELINE MODEL

- Logistic Regression
- Simple and Interpretable model
- Used as benchmark for comparison





BASELINE RESULTS

80%

Accuracy

0.77

Recall

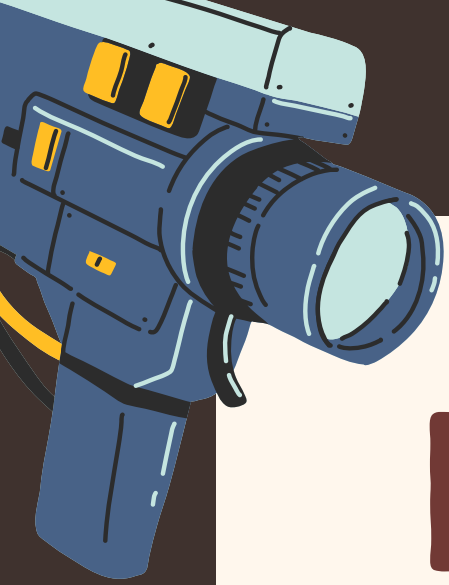
0.82

Precision

0.79

F-1 Score

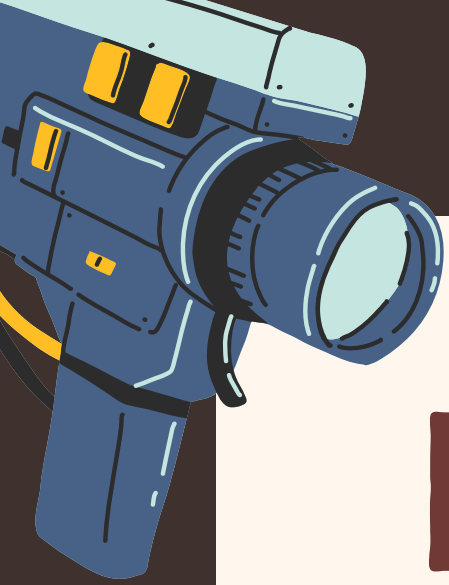




DEEP LEARNING MODEL

- Feedforward Neural Network
- Captures nonlinear relationships
- More flexible than logistic regression
- Learns complex patterns in movie data





NEURAL NETWORK RESULTS

82%

Accuracy

0.78

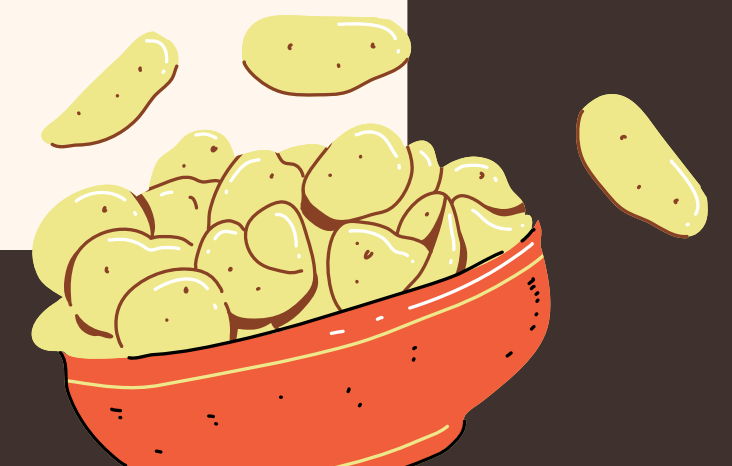
Recall

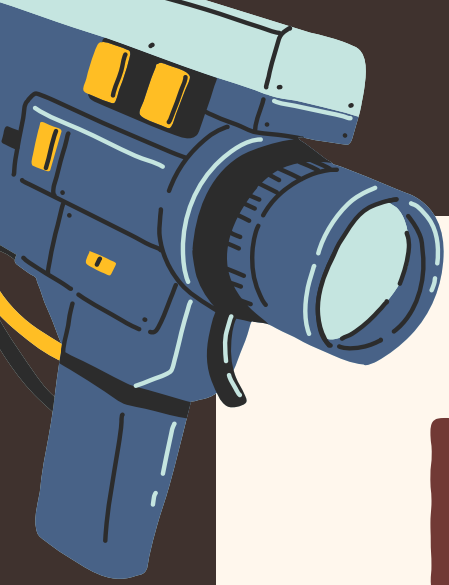
0.84

Precision

0.81

F1 Score

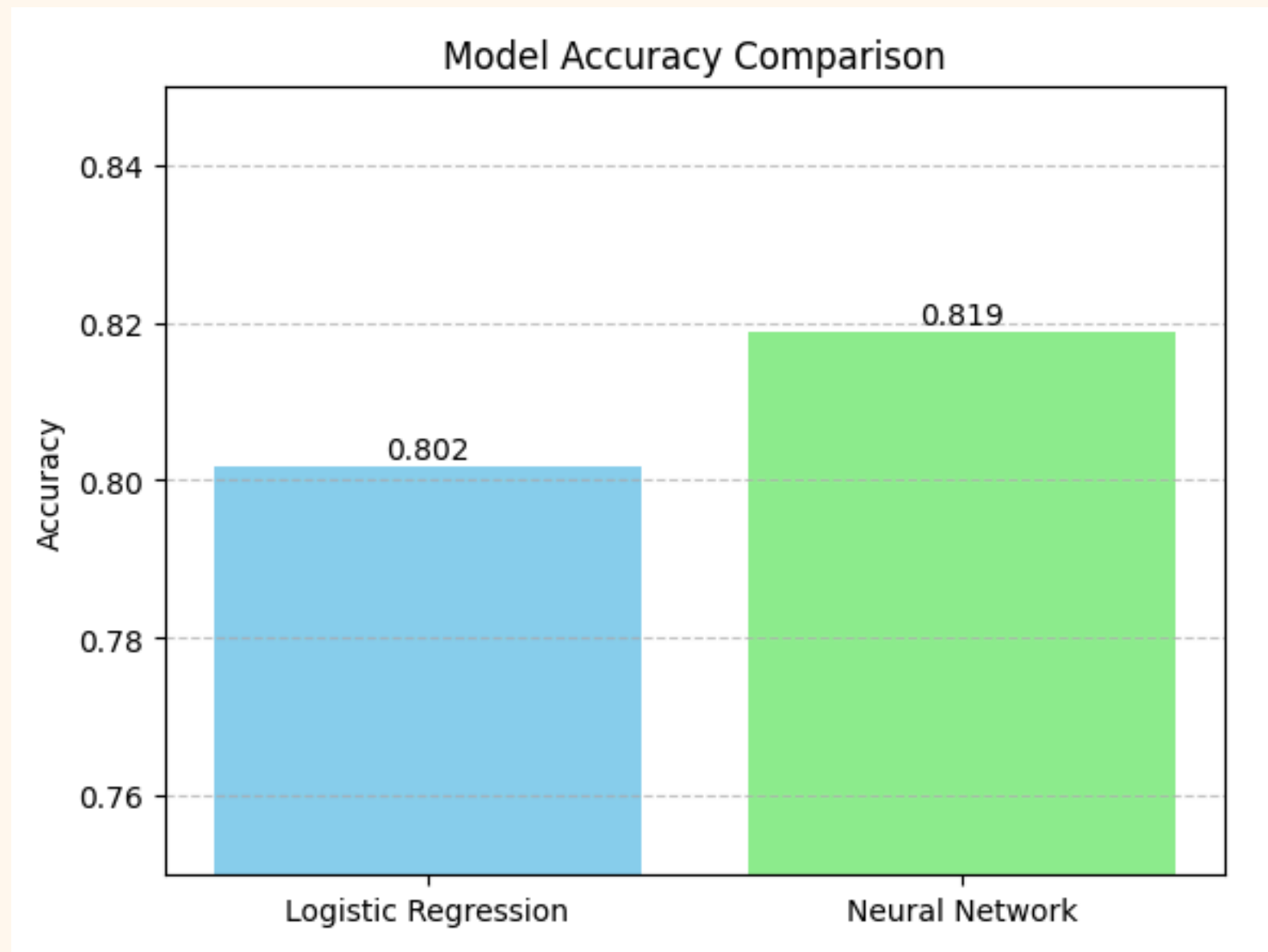


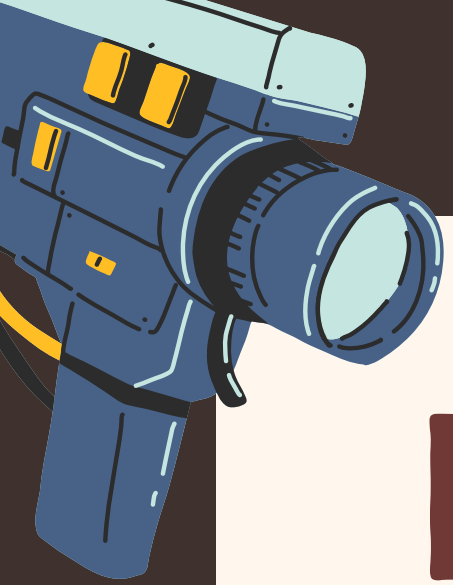


BASELINE VS DEEP LEARNING

Logistic Regression
Accuracy: 0.80

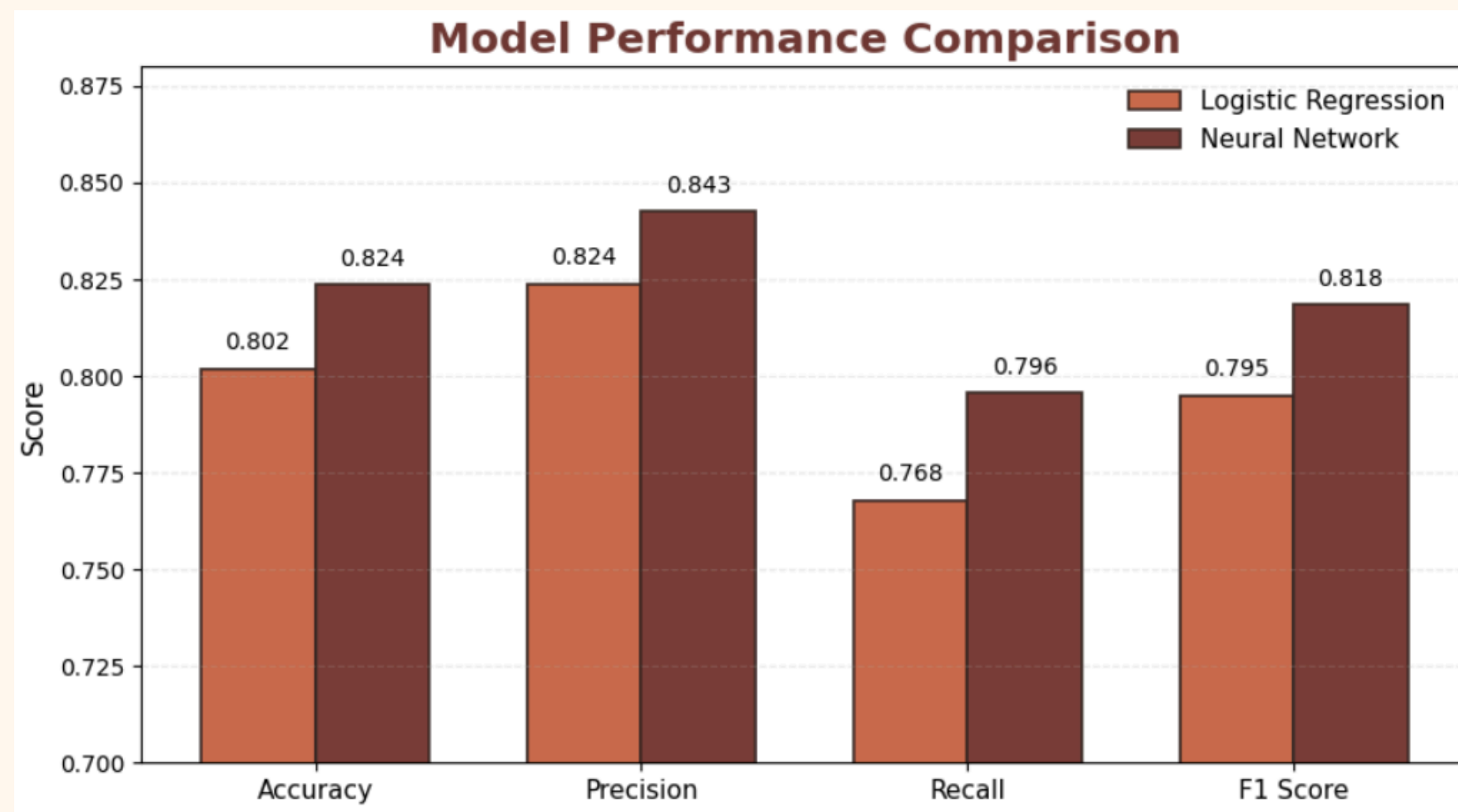
Neural Network
Accuracy: 0.82

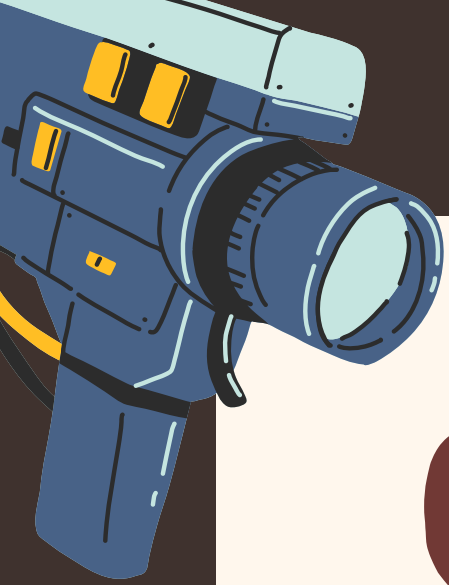




MODEL PERFORMANCE RESULTS

- Neural Network performed best across all metrics
- Accuracy improved from 0.802 to 0.824
- Baseline was already strong





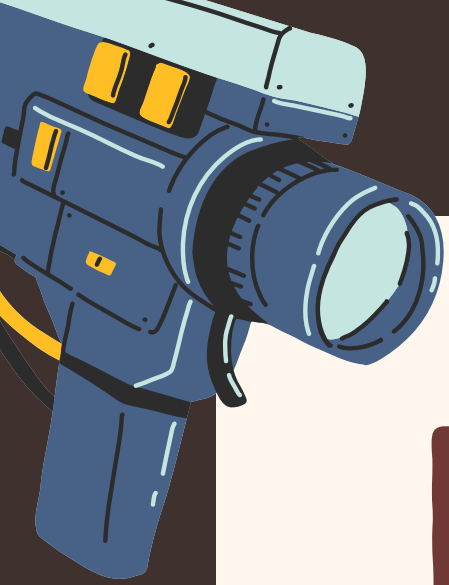
CONFUSION MATRIX INSIGHT

Confusion Matrix - Neural Network

Actual \ Predicted	0	1
0	275	48
1	66	257

- Correctly classified most movies
- 275 true negatives and 257 true positives
- Some successful movies were still missed
- Overall, predictions were balanced

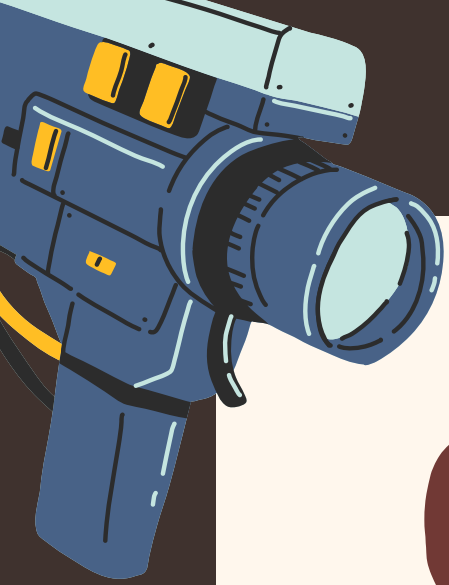




DISCUSSION

- Neural network improved over baseline
- Suggests movie success is not purely linear
- Better captures complex feature relationships
- Improvement was moderate, not dramatic

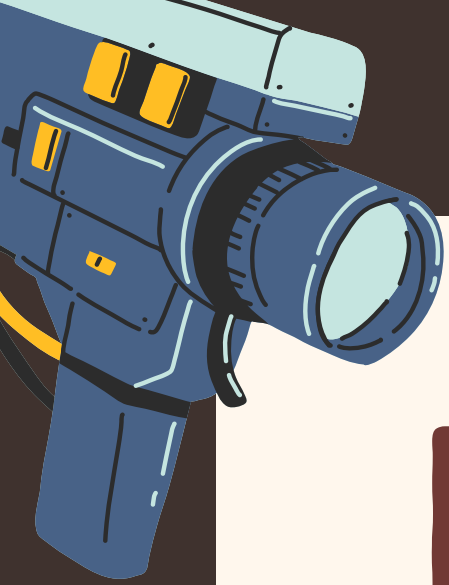




CONCLUSION

- Movie success can be predicted reasonably well
- Logistic regression was already a strong baseline
- Deep learning added value
- Final neural network accuracy: **~82.4%**





FUTURE WORK

- Add more features:
 - cast
 - marketing
 - release timing
- Test ensemble models
 - Random Forest / Gradient Boosting
- Use larger datasets





THANK YOU

For Your Attention